

Ali Parchekani

Toronto, ON, Canada

 (647) 766-7304

 aliparchekan@gmail.com

 [LinkedIn](#)

 [GitHub](#)

 [Google Scholar](#)

SUMMARY

Applied ML engineer and PhD researcher with 6+ years of experience across ML pipelines, optimization algorithms, and data-driven decision frameworks, with recent focus on agentic AI systems. Deep research background in statistical evaluation, experimentation, signal processing, and high-dimensional optimization, with hands-on engineering across LangGraph, OpenAI Agents SDK, FastAPI, Next.js, PyTorch, and MLOps tooling. Proven track record of translating complex research and product requirements into reproducible, evaluation-driven, deployable systems.

EXPERIENCE

Research Assistant - Applied Machine Learning and Wireless Sensing

January 2021 – Present

University of Toronto, Wireless Internet Research Lab, Toronto, ON

- Solved high-dimensional discrete phase-control problems in wireless sensing by designing a suite of optimization methods in Python and MATLAB — evolutionary search, Swendsen-Wang Markov chain Monte Carlo, coordinate descent, gradient-based solvers, and convex-relaxation baselines — and benchmarking them head-to-head to identify the strongest approach for each regime.
- Engineered end-to-end automotive radar simulation pipelines in Python using NumPy, SciPy, Matplotlib, and MATLAB Engine, spanning synthetic data generation, vehicle-to-vehicle channel modeling, range-Doppler processing, angle-velocity analysis, direction-of-arrival estimation, target detection, tracking, and multi-radar fusion — giving the lab a controlled testbed to validate interference-mitigation and deception methods before any hardware was involved.
- Made the simulation suite fully configurable through YAML/JSON-style experiment configurations, letting a single file vary transmitter and receiver geometry, vehicle trajectories, target states, subcarrier spacing, carrier frequency, noise power, interference-to-noise ratio, path loss, shadowing, fading, and intelligent-surface phase profiles.
- Wrote the signal-processing core — range estimation, Doppler and velocity estimation, angle estimation, false-alarm-controlled detection, radar-object tracking, and multi-frame trajectory evaluation — so baseline, optimized, and adversarial sensing systems could be compared on identical footing.
- Built a receiver-aware radar deception simulator that tunes intelligent-surface phase profiles to fabricate false target range, velocity, angle, and trajectory signatures while respecting radar scan timing, Doppler phase evolution, range-Doppler processing, and tracker-consistency constraints.
- Developed two-slot intelligent-surface control algorithms that suppress automotive radar interference by jointly optimizing phase configurations across consecutive sensing slots, while modeling mutual coupling, vehicle-channel dynamics, interference power, detection thresholds, and target-preservation constraints.
- Automated the full experimentation loop — batch simulation runs, metric logging, baseline comparisons, ablation and sensitivity studies, runtime benchmarking, and publication-ready tables and figures generated straight from experiment outputs — so results were reproducible and a paper's plots could be regenerated from a single command.
- Raised radar object-detection accuracy by 50% by designing RIS phase-optimization algorithms that suppress inter-radar interference in OFDM sensing.
- Cut training-data requirements by 20% and improved localization accuracy by 10% with model-based deep learning that bakes signal-processing structure directly into the network.
- Translated research results into system-design reports, simulation findings, and formal presentations for Huawei, informing the partner's R&D direction on wireless sensing, radar evaluation, and optimization-based intelligent-surface control.
- Published 4 peer-reviewed papers at IEEE VTC, ICC, and PIMRC, with a journal manuscript and conference paper under review and two more in preparation.

Data Scientist

January 2020 – January 2021

HARA AI, Tehran, Iran

- Built the natural-language-understanding layer for a Persian automated call-center system, converting speech-to-text transcripts into structured intents using text normalization, transformer-based classification, confidence scoring, and routing logic to trigger downstream customer-support actions.
- Constructed a 10M+ sentence Persian corpus from web crawls, news archives, and user-generated text; implemented cleaning, Unicode normalization, token standardization, deduplication, filtering, and train/validation dataset generation.
- Trained and fine-tuned transformer-based models in PyTorch, including BERT, ALBERT, and DistilBERT variants, with custom multi-class sequence-classification heads for Persian intent classification and customer-support query routing.

- Iterated on tokenizer design, sequence length, class-label structure, fine-tuning strategy, learning-rate schedules, and model-size/performance tradeoffs across multiple language-model variants.
- Built offline evaluation workflows — confusion-matrix analysis, class-level error inspection, and model-variant comparison — that drove intent classification to above 85% accuracy.
- Deployed trained models behind a FastAPI inference service feeding upstream speech-to-text and downstream dialogue management, delivering above 70% end-to-end accuracy in production.
- Owned the NLP classification stack from corpus construction and model training through evaluation, API integration, and deployment handoff in a cross-functional team.

Research Assistant - Network Traffic Analysis and Machine Learning

September 2018 – January 2021

University of Tehran, Wireless Multimedia Communications Lab, Tehran, Iran

- Developed machine-learning pipelines for network traffic classification, including virtual private network detection, application-level flow classification, and user-behavior modeling from packet-level and flow-level traffic records.
- Processed 500K+ network packet records using PySpark, pandas, NumPy, and Python, implementing data cleaning, flow aggregation, feature extraction, label preparation, and train/test dataset generation for large-scale traffic-analysis experiments.
- Engineered statistical and temporal traffic features, including packet-size distributions, inter-arrival times, flow duration, byte counts, packet counts, protocol attributes, and upstream/downstream traffic patterns.
- Designed and trained neural-network classifiers in Python using PyTorch, TensorFlow, Keras, NumPy, pandas, and scikit-learn, including convolutional, recurrent, and multilayer perceptron models for packet-level and application-level traffic classification.
- Evaluated model performance using classification accuracy, confusion matrices, class-level error analysis, and comparisons across neural-network architectures, feature representations, and traffic categories, achieving above 85% classification accuracy.
- Applied inverse reinforcement learning to sequential traffic data to infer latent user-behavior patterns and model goal-directed network activity without manually specified reward functions.
- Authored an arXiv-listed manuscript on neural-network-based virtual private network traffic classification with a rejection mechanism for low-confidence or ambiguous samples.

PROJECTS

Career Digital Twin - Serverless Conversational AI Agent

- Built a personalized conversational AI agent that answers questions about professional background, research, projects, publications, and experience using OpenAI models, structured profile data, and grounded knowledge files.
- Designed the system architecture with a serverless AWS backend using Lambda, API Gateway, CloudFront, and Amazon S3 for model orchestration, HTTP routing, static delivery, and knowledge-file storage, with a FastAPI inference layer and a Next.js frontend interface.
- Implemented infrastructure as code with Terraform, defining AWS resources, environment variables, IAM permissions, S3 buckets, API Gateway routes, Lambda deployment configuration, and CloudFront distribution settings.
- Developed prompt architecture with structured facts, profile summaries, grounding instructions, response-format constraints, and hallucination-reduction rules to produce accurate and professional answers across background-specific queries.

LangGraph Sidekick - Stateful Browser-Integrated AI Assistant

- Built a stateful multi-node LangGraph assistant with explicit graph state, persistent memory, conditional routing, tool-use approval gates, iterative self-correction, and separate planning, execution, evaluation, and response-generation nodes.
- Integrated browser automation, file handling, code execution, and structured tool-calling into a unified agent workflow, enabling the assistant to inspect web pages, manipulate local files, execute code, and revise outputs based on evaluator feedback.
- Extended the base architecture with human-in-the-loop approval for sensitive tool calls, expanded state schemas, custom workflow nodes, retry paths, and guardrails for safer execution of browser, file-system, and code-running operations.

Deep Research Multi-Agent System

- Designed a multi-agent research pipeline using the OpenAI Agents SDK and Gradio, with specialized planner, search, writer, and email agents connected through structured handoffs, typed outputs, and conditional routing logic.
- Implemented rate-limited parallel web search, query decomposition, result aggregation, source filtering, synthesis prompts, clarification checkpoints, and structured report generation for long-form research workflows.

Multi-Agent Engineering Team - CrewAI Software Builder

- Built a four-agent software-engineering workflow in CrewAI with manager, developer, reviewer, and tester agents coordinating through role-specific prompts, task decomposition, structured handoffs, and iterative feedback loops.
- Integrated Docker-based execution environments for generated code, enabling isolated build, run, test, and debugging cycles during autonomous software-development workflows.

- Designed agent responsibilities, review criteria, task memory, and validation steps so the system could plan application features, generate implementation code, review outputs, identify defects, and request corrections through multi-agent collaboration.

Short-Term Rental Price Prediction - Production-Style ML Pipeline

- Built a reproducible end-to-end machine-learning pipeline using scikit-learn, MLflow, Weights & Biases, DVC, and Hydra, covering data validation, feature engineering, train/validation/test splitting, random-forest training, hyperparameter tuning, and experiment tracking.
- Implemented versioned pipeline components with configuration-driven execution, artifact logging, model-metric tracking, dataset versioning, experiment comparison, and automated reporting for reproducible model-development workflows.

EDUCATION

Ph.D. in Electrical and Computer Engineering - Communications

January 2021 – Expected August 2026

University of Toronto, Toronto, ON

Thesis: Sensing and Scanning with Reconfigurable Intelligent Surfaces (in progress)

Candidacy achieved: Summer 2023

Relevant coursework: Algorithms for Collective Decision Making (A), Combinatorial Methods (A+), Non-Linear Optimization (A), Detection and Estimation Theory (A+), Game Theory and Evolutionary Games (A)

M.Sc. in Electrical and Computer Engineering - Communication Networks

September 2018 – December 2020

University of Tehran, Tehran, Iran

Thesis: Neural-Network-Based Network Traffic Classification and Aggregate User-Behavior Prediction Using Inverse Reinforcement Learning

GPA: 19.57/20 — ranked 1st in the department

Credential evaluation: Equivalent to a U.S. Master of Science in Electrical Engineering with emphasis in Communication Systems

B.Sc. in Electrical Engineering - Telecommunications

September 2014 – September 2018

University of Tehran, Tehran, Iran

GPA: 18.78/20 (3.98/4) — ranked 3rd in the department

PUBLICATIONS

Peer-Reviewed Conference Papers

1. A. Parchekani, S. Valaee, "Reconfigurable Intelligent Surface Assisted Sensing and Localization using the Swendsen-Wang and Evolutionary Algorithms," IEEE VTC 2023-Fall.
2. A. Parchekani, S. Valaee, "Sensing and Localization using Reconfigurable Intelligent Surfaces and the Swendsen-Wang Algorithm," IEEE ICC Workshops 2022.
3. A. Parchekani, S. Valaee, "Sensing via Orthogonal Time Frequency Space Signalling and Reconfigurable Intelligent Surface," IEEE PIMRC 2022.
4. A. Parchekani, M. Johnny, S. Valaee, "Model-Based Deep Learning Tuning of Reconfigurable Intelligent Surface for OFDM Radar Interference Mitigation," IEEE PIMRC 2025.

Preprints

1. A. Parchekani, S. Nouri, V. Shah-Mansouri, S. P. Shariatpanahi, "Classification of Traffic using Neural Networks by Rejecting: A Novel Approach in Classifying VPN Traffic," arXiv:2001.03665.
2. A. Parchekani, M. Johnny, S. Valaee, "Reconfigurable Intelligent Surface for OFDM Radar Interference Mitigation," arXiv:2502.14730.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, C, C++, MATLAB, LaTeX

Agentic AI and LLM Frameworks: LangGraph, OpenAI Agents SDK, CrewAI, MCP / Model Context Protocol, Microsoft AutoGen, OpenAI API, Anthropic API, Gradio, retrieval-augmented generation (RAG), embeddings, tool calling, structured outputs

ML and Deep Learning: PyTorch, scikit-learn, TensorFlow, Keras, Transformers

Scientific Computing and Data: NumPy, SciPy, Matplotlib, pandas, PySpark

Backend, APIs, and Web: FastAPI, Next.js, REST APIs, Pydantic, PostgreSQL, Redis, MongoDB

MLOps and Experiment Tracking: MLflow, Weights & Biases, DVC, Hydra, Docker, GitHub Actions, Git, Linux, Bash

AWS and Infrastructure: AWS Lambda, API Gateway, CloudFront, Amazon S3, EC2, VPC, IAM, RDS, Terraform

Signal Processing and Wireless: OFDM radar processing, direction-of-arrival estimation, range-Doppler analysis, target detection and tracking, vehicular channel modeling, intelligent surface phase optimization

Methods and Practices: Optimization, gradient descent, coordinate descent, convex relaxation, evolutionary algorithms, Markov chain Monte Carlo, statistical inference, experimental design, ablation studies, reproducible ML pipelines, CI/CD

ADDITIONAL HIGHLIGHTS

- Peer reviewer for IEEE GLOBECOM, ICC, PIMRC, and VTC, completing 5 technical reviews across wireless communications, machine learning, and signal processing venues from 2021 to 2025.
- Teaching assistant for 10+ University of Toronto courses across statistics, data science, machine learning, computer science, and mathematics, supporting hundreds of students from first-year through graduate level through tutorials, lectures, office hours, grading, and mentorship.
- Developed complete instructional packages for the University of Toronto International Summer Academy, including a 15-session AI and game theory course and a critical thinking in statistics course, with session plans, slides, assignments, project guides, and instructor materials.